4 **(a)** State the *first law of thermodynamics*.

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……

…… [1]

(b) A diesel engine containing an ideal gas at a pressure of 1.0×10^5 Pa, temperature of 300 K and volume of 10×10^{-4} m^3 undergoes a cyclic process in the sequence as follows:

A the gas is compressed adiabatically (i.e. no net heat transfer) to a pressure of 16×10^5 Pa, temperature of 660 K and volume of 1.4×10^{-4} m^3

B the gas is expanded at constant pressure to a volume of 6.0×10^{-4} m^3

C the gas continued to expand adiabatically to a pressure of 7.8×10^5 Pa to its original volume

D the gas is cooled at constant volume to its original temperature and pressure

(i) Illustrate these changes on the *p*-*V* diagram in Fig. 4.1.

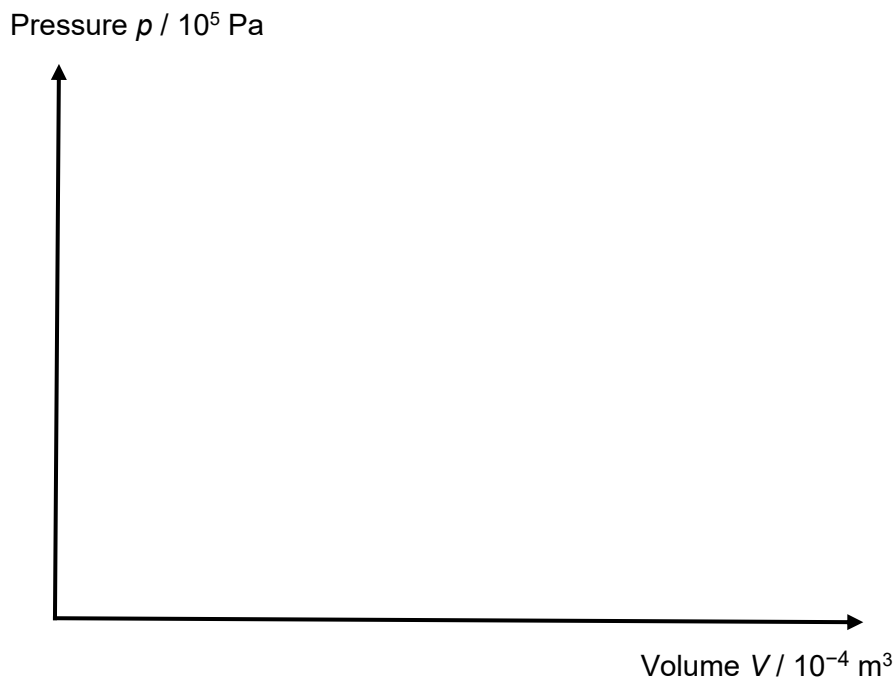

Fig. 4.1 [4]

(ii) Fig. 4.2 shows the changes in internal energy, heat supplied and work done on gas during the cyclic process. Complete Fig. 4.2.

	Heat supplied to gas / J	Work done on gas / J	Increase in internal energy of gas / J
A	0	300	300
B	2580		
C	0	−440	−440
D	−1700		

Fig 4.2

[3]

(iii) Assuming that the efficiency of the diesel engine is defined as ratio of net work done by the gas in the whole cycle to heat supplied to gas during process **B**, calculate the efficiency of this engine.